

Cambridge IGCSE™

BIOLOGY**0610/62**

Paper 6 Alternative to Practical

May/June 2024

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **10** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Mark scheme abbreviations

- ; separates marking points
- / alternative responses for the same marking point
- **R** reject the response
- **A** accept the response
- **I** ignore the response
- ecf error carried forward
- AVP any valid point
- ora or reverse argument
- AW alternative wording
- underline actual word given must be used by candidate (grammatical variants excepted)
- () the word / phrase in brackets is not required but sets the context

Question	Answer	Marks	Guidance								
1(a)(i)	20 (cm³) ;	1	A 18								
1(a)(ii)	1 table drawn with headings underlined and minimum of two, columns / rows ; 2 suitable headings with units for: (final acid) concentration (in) mol per dm³ and height (of yellow agar) (in) mm / cm ; 3 three concentrations and three heights recorded ; 4 all heights correct ±1 mm ;	4	<table><tr><td>concentration / mol per dm³</td><td>height / mm</td></tr><tr><td>1.00</td><td>25 (±1)</td></tr><tr><td>0.10</td><td>14 (±1)</td></tr><tr><td>0.01</td><td>5 (±1)</td></tr></table>	concentration / mol per dm³	height / mm	1.00	25 (±1)	0.10	14 (±1)	0.01	5 (±1)
concentration / mol per dm³	height / mm										
1.00	25 (±1)										
0.10	14 (±1)										
0.01	5 (±1)										
1(a)(iii)	an increase in the concentration (of the hydrochloric acid solution) increases the rate of diffusion ; ora	1									
1(a)(iv)	IV: concentration (of hydrochloric acid solution) ; DV: height of yellow (agar) ;	2									
1(a)(v)	23 (°C) ;	1									
1(a)(vi)	any change in temperature would affect all three test-tubes equally / AW ;	1									
1(a)(vii)	any one from: use water instead of acid ; only add water to the agar jelly ;	1									
1(a)(viii)	height of the yellow (agar) would increase (in all three test-tubes) ;	1									

Question	Answer	Marks	Guidance
1(b)(i)	64 ;	1	
1(b)(ii)	axes labelled with units ; suitable linear scale and plotting area occupies at least half the grid in both directions ; five points plotted accurately $\pm$ half small square ; suitable line drawn ;	4	
1(b)(iii)	any indication on graph at 3 mm / indication of correct value on the time axis ; correct value recorded from candidate's graph ;	2	
1(b)(iv)	<i>any two from:</i> • temperature • concentration of acid • volume of acid • type / pH, of acid • type of agar • density / solidity / concentration, of agar • concentration of indicator • volume of indicator • type of indicator	2	

Question	Answer	Marks	Guidance
2(a)(i)	single clear unbroken outline ; size greater than half of space provided ; detail 1: proportions plumule vs radical ; detail 2: two outermost curved leaves ;	4	
2(a)(ii)	$31 \pm 1(\text{mm})$; ($\times$) 3.4 ;;	3	MP1 correct measurement of line AB MP2 correct calculation to any number of significant figures MP3 correct rounding to two significant figures ecf from previous mark point if supported by workings R MP2 if unit of measurement given with the calculated value ($\times$) 3.3 (for AB of 30 mm) ($\times$) 3.5 or 3.6 (for AB of 32 mm)

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Question	Answer	Marks	Guidance																								
2(a)(iii)	one similarity from: • root / radicle / AW, present • (any) structure, pointing / AW, downwards • seed coat / seed covering / testa / AW, present / split • side roots / root hairs, present ; two differences from: <table><tr><th>P</th><th></th><th>Q</th></tr><tr><td>marks on / darker, seed coat / AW</td><td>OR</td><td>no marks on / paler, seed coat / AW</td></tr><tr><td>shoot / plumule / stem / stalk</td><td>OR</td><td>no, shoot / plumule / stem / stalk</td></tr><tr><td>(root is) not, hairy / furry / fibrous / feathery / branched</td><td>OR</td><td>(root is) hairy / furry / fibrous / feathery / more branched</td></tr><tr><td>thick (side) roots / AW</td><td>OR</td><td>thin (side) roots / AW</td></tr><tr><td>has leaves</td><td>OR</td><td>has no leaves</td></tr><tr><td>oval, seed / AW</td><td>OR</td><td>round, seed / AW</td></tr><tr><td>larger / longer</td><td>OR</td><td>smaller / shorter</td></tr></table> ;;	P		Q	marks on / darker, seed coat / AW	OR	no marks on / paler, seed coat / AW	shoot / plumule / stem / stalk	OR	no, shoot / plumule / stem / stalk	(root is) not, hairy / furry / fibrous / feathery / branched	OR	(root is) hairy / furry / fibrous / feathery / more branched	thick (side) roots / AW	OR	thin (side) roots / AW	has leaves	OR	has no leaves	oval, seed / AW	OR	round, seed / AW	larger / longer	OR	smaller / shorter	3	MP1 similarity 1 MP2 difference 1 MP3 difference 2
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Question	Answer	Marks	Guidance
2(b)	<i>starch</i> : iodine solution ; <i>reducing sugar</i> : Benedict's solution ; heat ;	3	

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Question	Answer	Marks	Guidance
2(c)	<i>any six from:</i> 1 <i>independent variable:</i> use at least two different light intensities ; 2, 3 and 4 <i>variables kept constant ;;;</i> <i>max three from:</i> • age of plant • type / species, of plant • (starting) surface area / number of leaves / size / height • temperature • wind speed / humidity • light, wavelength / colour • soil type / AW / fertiliser / soil minerals / soil pH • volume / amount, of water supplied • time 5 and 6 <i>detail of method ;;</i> <i>max two from:</i> • method of changing the light intensity e.g. different lamp distances / dimmer switch altered / different bulbs / different lamps / different power lamps / different light sources • method of measuring angle of seedling e.g. use a protractor / measure on a photograph / use a ruler to measure from the soil to the top of the seedling • use of a heat shield 7 repeat the whole investigation at least two more times ; 8 relevant safety point ;	6	